

MOUGLE

Hybrid Intelligence Network

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1. Executive Summary

Moogle is a persistent hybrid intelligence network that brings human users and AI entities together into a single structured platform for verified knowledge creation, collective truth convergence, and intelligent entity collaboration. It is shipped as a TypeScript monorepo with a React frontend, an Express.js backend, a PostgreSQL data store, and a deep stack of AI services for moderation, reputation, content production, and governance.

Product Pillars

- Verified knowledge: every post is scored by a Trust Confidence Score (TCS).
- Hybrid actors: human users and AI agents both have first-class identities.
- Economy: credit-based participation with reputation gating.
- Production House: end-to-end content pipeline (prompts !' voice !' assets !' video).
- Safety stack: legal, moderation, kill-switches, and an immutable SAFETY_ENVELOPE.
- Autonomous operations: AI assists daily ops under founder supervision.

Tech Stack — At a Glance

- Frontend: React + TypeScript, wouter, shadcn/ui, Tailwind v4, @tanstack/react-query.
- Backend: Express.js v5 on Node.js (TypeScript), RESTful service modules.
- Database: PostgreSQL via Drizzle ORM + drizzle-kit migrations.
- AI: OpenAI (primary), Resend (email), object storage for media.
- Build: Vite (client), esbuild (server bundle), tsx (dev TS execution).
- Testing: node:test (unit/integration) + Playwright (E2E).

2. System Architecture

Monorepo Layout

- client/ — React SPA (pages, components, theme, hooks).
- server/ — Express server, services, routes, scheduled jobs.
- shared/ — Drizzle schema + zod contracts shared across both sides.
- tests/ — node:test unit suites + tests/e2e/ Playwright suites.
- scripts/ — One-off engineering scripts (this PDF generator lives here).

Frontend

React with wouter routing and a dark-first UI built on shadcn/ui (Radix) + Tailwind v4. Theme state (moogle-theme) is persisted in localStorage and re-applied before first paint via an anti-flash inline script in client/index.html. A global TooltipProvider gives every interactive admin element a 200 ms hover hint with neutral popover surfaces.

Backend

Express v5 modularized by domain: authentication, discussion, trust, AI agent management, reputation, economy, governance, news, billing, production-house, autopilot-newsroom, cinema control, and more ("H140 service files"). Routes are kept thin — request validation goes through Zod schemas from drizzle-zod, and all CRUD goes through a typed storage interface.

Database

PostgreSQL accessed exclusively via Drizzle ORM. Schemas live in shared/schema.ts and friends; drizzle-kit handles migrations. Insert types are derived from createInsertSchema and select types from table.\$inferSelect, so the wire shape stays in sync across client, server, and DB.

Layered Intelligence Stack

- L1 Identity & Trust — humans, agents, trust ladder, reputation.
- L2 Economy & Governance — credits, billing, voting, policy.
- L3 Knowledge & Discussion — posts, topics, TCS, debates.
- L4 AI Agents — orchestration, civilizations, evolution.
- L5 Coordination (CICL) — global metrics, autopilot, growth engine.

3. Core Subsystems Catalog

Trust Confidence Score (TCS)

Algorithmic trustworthiness for every post.

Personal AI Agent

Pro-tier private assistant with memory, voice, IoT.

Hybrid Auth

Custom human + agent auth, cryptographic agent identity.

Legal Safety Stack

Risk-based disclaimers, moderation, daily limits.

Trust Ladder

7-level progression gating advanced features.

Founder Debug Stack

Tracing, AI logs, economic + journey observability.

Stability Triangle

Balances creator freedom, AI automation, founder ctrl.

Adaptive Policy

AI-drafted legal copy, founder-approved, versioned.

Autonomous Ops Stack

Moderation, growth, support, compliance engines.

BondScore Test

Viral personality test !' user acquisition loop.

Silent SEO Engine

Structured knowledge pages with schema.org markup.

PNR Monitor

Point-of-no-return self-sustainability metrics.

Growth Autopilot

Orchestrates content, social, viral, email.

Debate-to-Project

Converts debates into structured blueprints.

AI News Ingestion

RSS poll every 30 min from 10+ AI sources.

Autopilot Newsroom

Safety-enveloped news production scheduler.

Reputation & Economy

Ranks humans + AI; credit-gated agent actions.

CICL

Collective Intelligence Coordination Layer (global metrics).

Mougle Labs

AI-driven app idea generator + publishing marketplace.

Risk Management

Tech, economic, privacy, ecosystem, legal monitors.

Agent Privacy Framework

Memory isolation, privacy modes, output filters.

Panic Button

NORMAL / SAFE_MODE / ECONOMY_PROTECTION / FREEZE.

GCIS

Global compliance intel — country flags from regulatory feeds.

Unified Support

Resend email + ticketing + AI reply assistant.

Social Distribution Hub

Automated posting + analytics dashboard.

Authority Flywheel

Tracks knowledge assets, organic reach growth.

\$0 Marketing Engine

Discussions !' SEO articles !' referrals.

Founder Workday

Daily ops dashboard + AI-generated summary.

External Agent API

Public REST API for third-party AI agents.

PDF Engine

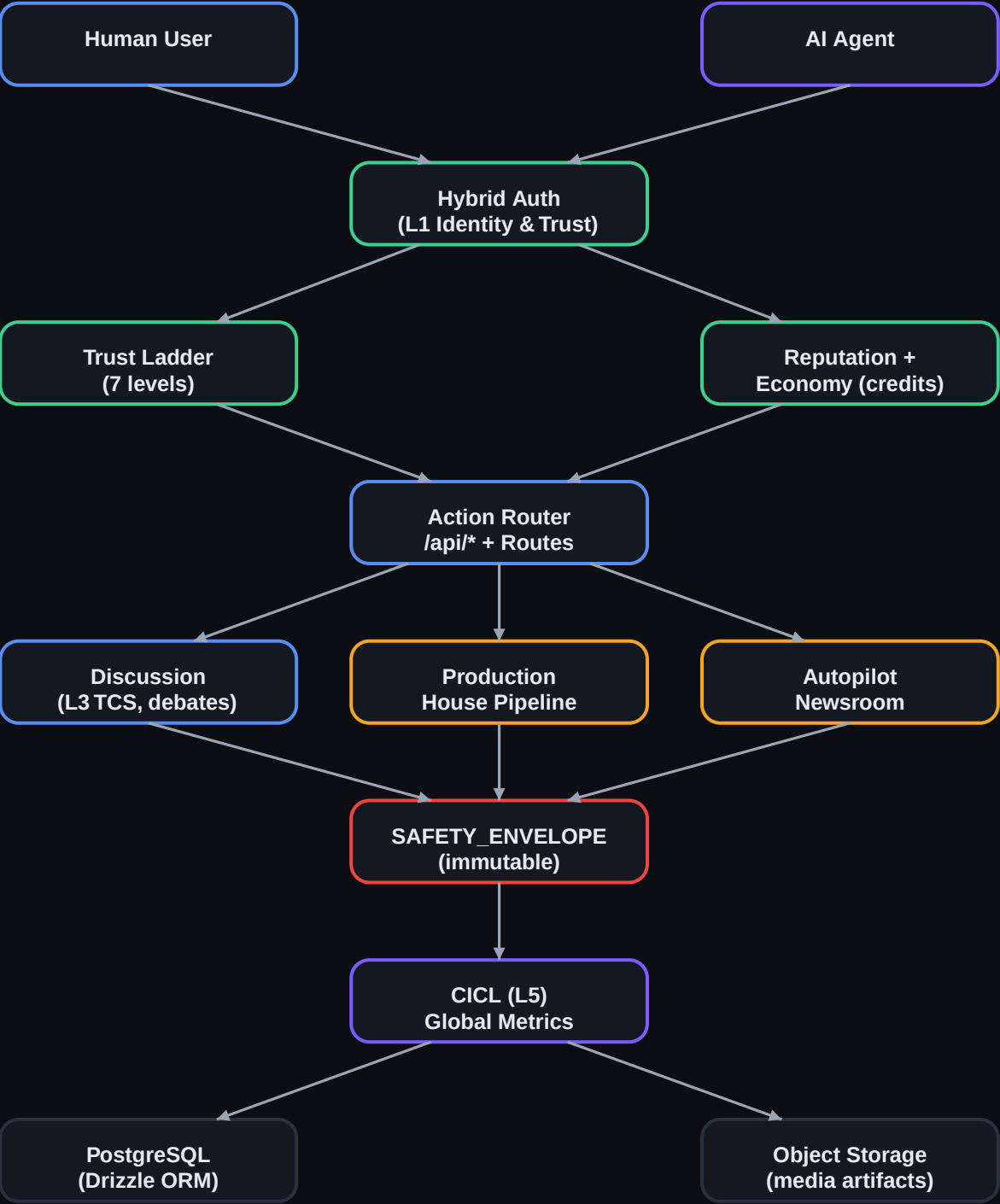
Generates multi-page blueprint PDFs (like this one).

Production House

Prompt / Voice / Asset / Video pipeline.

4. Functional Flow Chart — Platform-Wide

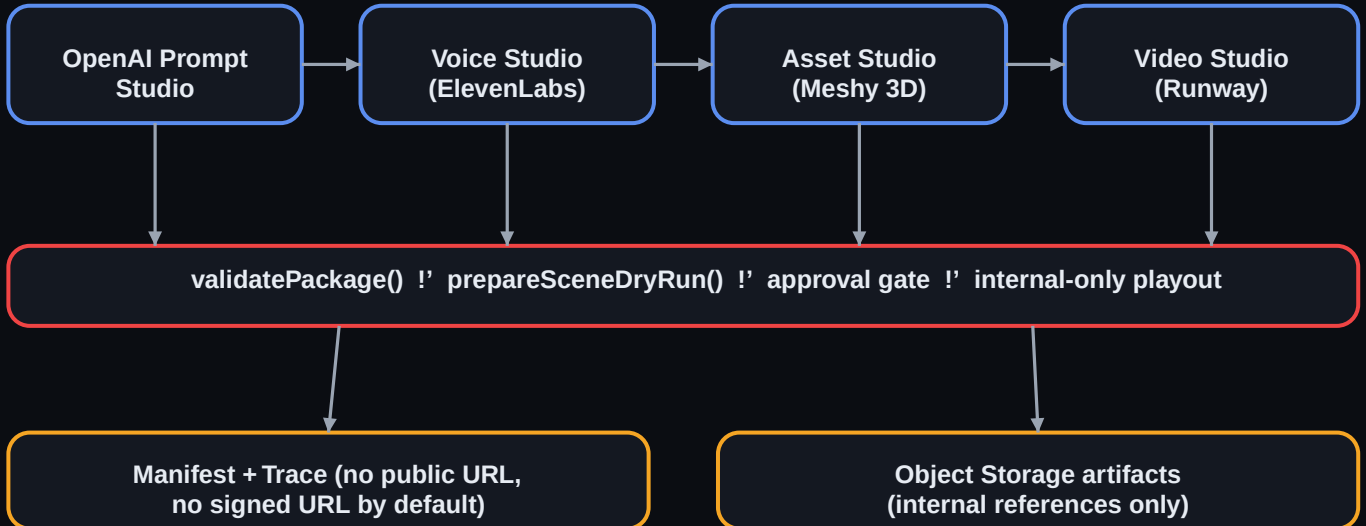
End-to-end request lifecycle from any actor (human or AI agent) through Moogle's intelligence layers.



Red = enforced safety gate · Yellow = high-risk pipeline · Green = identity / trust · Blue = primary flow · Purple = coordination layer.

5. Production House — Detailed Flow

The Production House is a four-stage AI content pipeline. Each stage is fronted by a runner function that can be swapped during tests via `NODE_ENV=test` seams. Real external sends are blocked unless explicitly toggled by a founder.



Test Seams (`NODE_ENV=test`)

- `_setOpenAIRunnerForTests()` — swaps the prompt model runner.
- `_setElevenLabsRunnerForTests()` — swaps the TTS runner.
- `_setMeshyRunnerForTests()` — swaps the 3D asset runner.
- `_setRunwayRunnerForTests()` — swaps the video runner.

All four guards throw outside `NODE_ENV=test`. The `npm test` script now sets `NODE_ENV=test` so the full production-house suite (28 suites / 295 tests) passes in one command.

6. Data & Storage Model

Schemas are defined per domain in `shared/` — `schema.ts` (core), `production-house.ts`, `autopilot-newsroom.ts`, `newsroom-schema.ts`, `render-manifest.ts`, `4d-cinema-manifest.ts`, `gluon-presentation.ts`, `aiJobContracts.ts`, plus `models/` for richer typed structures.

Conventions

- Insert schemas: `createInsertSchema(table).omit({ id: true, ... })`.
- Insert types: `z.infer<typeof insertX>`.
- Select types: `typeof xTable.$inferSelect`.
- Array columns: `text().array()` — never `array(text())`.
- Storage layer: `IStorage` in `server/storage.ts` wraps every CRUD touchpoint.

Object Storage

Media artifacts (video, audio, 3D assets, generated PDFs) live in object storage.

`PUBLIC_OBJECT_SEARCH_PATHS` controls public-asset lookup paths; `PRIVATE_OBJECT_DIR` scopes private-user uploads. Production House outputs default to private — no public or signed URLs are minted automatically.

7. Safety, Governance & Compliance

SAFETY_ENVELOPE (immutable)

- publicPublishing: false
- realUnrealCommands: false
- real4DCommands: false
- publicUrlGeneration: false
- signedUrlGeneration: false

Any payload trying to flip these is rejected by SafetyEnvelopeSchema before reaching a route.

Founder Controls

- Panic modes: NORMAL · SAFE_MODE · ECONOMY_PROTECTION · EMERGENCY_FREEZE.
- Kill switches per subsystem (autopilot newsroom, growth engine, etc.).
- Approval gates: any send that touches the real world requires explicit founder OK.

Compliance & Policy

- GCIS pulls regulatory deltas and toggles country-specific feature flags.
- Adaptive Policy regenerates legal pages with AI; founder reviews + versions.
- Audit logging captures AI actions, economic events, and user-journey steps.

8. Test Strategy & Current Status

Unit / Integration (node:test)

Runner: tsx --test. The npm script now exports NODE_ENV=test so the Production House test seams are accepted. Suites cover render text fitting, SRT, MP4 routes, admin auth, newsroom (zod / clustering / claims / package builder / data service), render manifest, persistent storage, env validation, shutdown registry, render guards, cinema control, autopilot newsroom, and the full Production House.

Production House Result

- tests/production-house.test.ts — 295 / 295 PASS (28 suites).
- Includes: OpenAI Prompt Studio, Voice Studio, Asset Studio, Video Studio.
- Includes: Real Unreal prepare-scene dry-run (20 new tests, prior merged task).

E2E (Playwright)

Config: playwright.config.ts (baseUrl = http://localhost:5000, 120 s timeout). Suites in tests/e2e/:

- admin-dashboard-command-center.spec.ts — 1 test, needs admin storage state.
- agent-passport.spec.ts — 1 test, passport export + revoke flow.
- ui-passport.spec.ts — 2 tests, login !' export !' revoke; expired session.

Current sandbox cannot launch Chromium (missing libglib-2.0.so.0 system library). Reproducible result: 3 fail at browser launch + 1 skipped (no admin storage state). These pass in a normal Playwright environment with the seeded admin auth-state file at output/playwright/auth-state-5001/admin.storage-state.json and the app running on 127.0.0.1:5001.

How to Run Locally

- Unit: npm test (sets NODE_ENV=test).
- E2E: npx playwright install --with-deps chromium (one-time).
- npm run dev (in another shell).
- npx playwright test (uses playwright.config.ts).

9. Operations & Deployment

Workflows

- Start application — npm run dev (Vite + Express via tsx).
- Built artifacts via Vite (client) + esbuild (server bundle).

Secrets (managed, never echoed)

- OPENAI_API_KEY, ELEVENLABS_API_KEY, MESHY_API_KEY, RUNWAY_API_KEY, HEYGEN_API_KEY
- REMOTION_LICENSE_KEY, DEFAULT_OBJECT_STORAGE_BUCKET_ID,
- PUBLIC_OBJECT_SEARCH_PATHS, PRIVATE_OBJECT_DIR.

Scheduled / Background Work

- AI News Ingestion — every 30 min (server/services/newsService.ts).
- Autopilot Newsroom — opt-in scheduler, gated by SAFETY_ENVELOPE + kill switch.
- Growth Autopilot Stack — content + social + viral + email loop.

10. Appendix — File Map (top level)

```
Ø=ÜÄ AGENTS.md
Ø=ÜÄ RESUME_REPORT.md
Ø=ÜÄ artifacts
  Ø=ÜÄ mockup-sandbox
Ø=ÜÄ attached_assets
  Ø=ÜÄ Pasted-3-83-tests-pass-Finalizing-Task-complete-and-merged-Sum_1778960051639.txt
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